Top Biases by Citation Count
(Bubble size proportional to citation count, showing biases with count ≥ 10)

Omics Category

Genomics
- Selection/ascertainment
- Ancestry Bias
- Variant Filtering
- Algorithmic Bias

Transcriptomics
- Low-Expression Bias
- Dropout / Sparsity
- Amplification Bias
- RNA Degradation
- Integration Correction
- Normalization Issues
- Cost / Resources
- Spatial Resolution

Metabolomics
- Pre-analytical Variability
- Cohort Confounders
- Matrix Effects
- Batch Effects
- Feature Detection
- Batch Correction
- Incomplete Metadata
- Reproducibility
- Standardization Gaps
- Cost / Resources

Proteomics
- Sensitivity & Specificity
- Data Processing Challe...

Multi-omics
- Sample Degradation
- Sample Heterogeneity
- Platform Limits
- Amplification Bias
- High-Dim / Overfitting
- Data Integration
- Black-Box Models
- Reproducibility
- Siloed Integration
- Standardization Gaps

General Omics
- Temporal Data Gaps
- Sample Heterogeneity
- Sample Prep Bias
- Batch Effects
- Database Gaps
- Pipeline Heterogeneity
- Privacy & Regulation
- Standardization Gaps

Chinese Literature
- Participation Limits
- Underrepresentation
- Allelic Dropout
- Short-read Limitations
- Haplotype Phasing
- Database Gaps
- Resource Inequality
- Ethical Barriers

Citation Count: 10, 15, 20, 25, 30

Bias Type (Subcategory)

Data Production | Technical/Instrumental | Computational/Analytical | Reporting/Interpretation | Other Challenges